TERTIARY INFOTECH ACADEMY PTE LTD

UEN: 201200696W

LESSON PLAN

For

Semiconductor Device and Process Integration Masterclass

Course Code: C605

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 Oct 2026	First version	Tertiary Infotech Academy

TABLE OF CONTENTS

Course Overview and Learning Outcomes 4

Day 1 Schedule - Device Physics and Fabrication (9:30am – 5:30pm) 4

Day 2 Schedule - Integration, Verification and Action (9:30am – 5:30pm) 5

Topic-by-Topic and Activity Facilitation Maps 6

Resources 8

Course Overview

A two-day, 15-hour course of classroom facilitation, running 9:30am – 5:30pm each day (7.5 instructional hours per day). The learning sequence moves from device physics and fabrication architecture to process interactions, performance verification, reliability and follow-up action.

Learning Outcomes

- LO1: Identify semiconductor manufacturing process flows based on device physics.
- LO2: Determine follow-up actions for process integration issues and performance.

Day 1 Schedule - Device Physics and Fabrication (9:30am – 5:30pm)

Time	Duration	Topic / Activity	Method	Slides
09:30-10:00	30 min	Welcome, introductions, ground rules, course outline and lesson plan	Facilitation	1-8
10:00-10:55	55 min	Semiconductor concepts and Activity 1 briefing	Lecture / questions	9-31
10:55-11:30	35 min	Activity 1: Device Physics Evidence Map	Individual / peer review	416
11:30-12:20	50 min	PN junctions, diodes, MOS capacitors, MOSFETs and BJTs	Lecture / worked examples	32-97
12:20-12:50	30 min	Lunch	Break	-
12:50-13:55	65 min	Memory, wafer preparation, oxidation and RTP	Lecture / peer sharing	98-130
13:55-14:35	40 min	Activity 2: Process Module Architecture	Individual / peer review	417
14:35-15:45	70 min	Lithography, CMP, etch, doping, deposition and metallisation	Lecture / case prompts	131-241
15:45-16:30	45 min	CMOS flow and design rules	Lecture / cross-section review	242-280
16:30-17:20	50 min	Activity 3: CMOS Manufacturing Flow and Control Gates	Individual / peer review	418
17:20-17:30	10 min	Day 1 synthesis and Day 2 preview	Facilitation	10, 281

Day 2 Schedule - Integration, Verification and Action (9:30am – 5:30pm)

Time	Duration	Topic / Activity	Method	Slides
09:30-09:50	20 min	Day 1 retrieval	Facilitation	281
09:50-10:45	55 min	Integration interactions and yield concepts	Lecture / discussion	281-322
10:45-11:30	45 min	Activity 4: Lithography-Etch Interaction Case	Case study / peer challenge	419
11:30-12:15	45 min	Doping and thermal evidence revisit	Lecture / worked evidence	203-221
12:15-12:45	30 min	Lunch	Break	-
12:45-13:30	45 min	Activity 5: Doping and Thermal-Budget Interaction	Case study / peer challenge	420
13:30-14:05	35 min	Performance and reliability evidence	Lecture / failure analysis	323-366
14:05-15:00	55 min	Activity 6: Yield and Performance Metric Verification	Case study / peer challenge	421
15:00-15:35	35 min	Packaging, final test and emerging integration	Lecture / examples	367-415
15:35-16:20	45 min	Activity 7: Reliability Failure Analysis	Case study / peer challenge	422
16:20-17:20	60 min	Activity 8: Follow-Up Action and Integration Review	Case study / peer challenge	423
17:20-17:30	10 min	Course summary, feedback and close	Facilitation	424-427

Topic-by-Topic Facilitation Map

LU1: Fundamentals of Device Physics and Device Fabrication - Slides 9-280

- Semiconductor materials and carrier physics - Slides 9-31
- PN junctions and diode behaviour - Slides 32-43
- MOSFET and BJT device operation - Slides 44-97
- Memory devices and integrated circuits - Slides 98-111
- Wafer fabrication process modules - Slides 112-241
- CMOS integration flows - Slides 242-275
- Layout and design rules - Slides 277-280

LU2: Process Integration Issues - Slides 281-415

- Isolation, wells and self-aligned modules - Slides 281-297
- Contamination, yield and process variation - Slides 298-346
- Device performance characterisation - Slides 347-355
- Device reliability and failure mechanisms - Slides 356-366
- Assembly, packaging and final test - Slides 367-399
- Optimisation and emerging technologies - Slides 400-415

Activity Facilitation Map

Activity 1: Device Physics Evidence Map - Slide 416

Purpose: Connect device-physics mechanisms to observable manufacturing controls and device consequences.

- Method: individual analysis followed by peer challenge; planned duration 35 minutes.
- Evidence: Completed device-physics evidence map with at least six mechanism-control-consequence links.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 2: Process Module Architecture - Slide 417

Purpose: Identify the major process loops, their purposes, inputs, outputs and control gates.

- Method: individual analysis followed by peer challenge; planned duration 40 minutes.
- Evidence: Process architecture map with module boundaries, control gates and two feedback loops.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 3: CMOS Manufacturing Flow and Control Gates - Slide 418

Purpose: Outline a coherent CMOS process flow and show where device physics constrains sequencing.

- Method: individual analysis followed by peer challenge; planned duration 50 minutes.

- Evidence: One-page CMOS process flow with control gates and three device-physics sequence rationales.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 4: Lithography-Etch Interaction Case - Slide 419

Purpose: Determine how variation propagates between lithography, etch, clean and contact modules.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Interaction matrix, causal chain and discriminating verification plan.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 5: Doping and Thermal-Budget Interaction - Slide 420

Purpose: Analyse implantation, activation, diffusion and oxide interactions across the thermal budget.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Thermal-budget comparison, evidence map and split-lot experiment design.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 6: Yield and Performance Metric Verification - Slide 421

Purpose: Select and interpret metrics that verify process-integration performance.

- Method: individual analysis followed by peer challenge; planned duration 55 minutes.
- Evidence: Metric dashboard worksheet with calculations, spatial interpretation and verification conclusion.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 7: Reliability Failure Analysis - Slide 422

Purpose: Verify whether TDDB, hot-carrier degradation or electromigration best explains the evidence.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Failure-mechanism comparison and reliability verification statement.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 8: Follow-Up Action and Integration Review - Slide 423

Purpose: Determine proportionate containment, corrective, preventive and verification actions.

- Method: individual analysis followed by peer challenge; planned duration 60 minutes.
- Evidence: Integration review action register with containment, CAPA, owners and effectiveness checks.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Resources Required

- Current trainer slides and learner slides
- Learner Guide
- Individual activity folders and printable PDFs
- Laptop with spreadsheet/PDF viewer
- Projector or display, whiteboard and markers
- LMS access

LMS: https://lms-tms.tertiaryinfotech.com/